

Skills Summary

Subtracting Across a Zero

Use this reference while completing your worksheet or when studying.

Skill 1 — Borrowing when the tens are zero

Rule: A column that cannot give enough borrows from the column on its left.

A zero has nothing to lend. Go one more column left: open a hundred into ten tens first, and only then lend a ten to the ones.

Every time a column lends, its own digit goes down by one. Cross it out and write the new digit above.

Example: Subtract $405 - 132$.

Step 1. The ones are fine here, but the tens column is empty and still has to give three tens away, so the borrowing has to start one column further left than usual.

Step 2. Open one hundred into ten tens: three hundreds are left and the tens column now holds ten. Ones: five take two leaves three. Tens: ten take three leaves seven. Hundreds: three take one leaves two.

273

Try it: Subtract $807 - 264$.

check: **543**

Skill 2 — Subtracting from hundreds with two zeros

Rule: When both the tens and the ones are zero, regroup once across the whole number before subtracting anything.

A round hundreds number becomes: one hundred fewer, nine tens, and ten ones. Then every column has something to give.

Example: Subtract $300 - 148$.

Step 1. Neither the ones nor the tens can give anything away, so rather than borrow twice in the middle of the work I rewrite the whole number in a form where every column is stocked.

Step 2. Three hundreds becomes two hundreds, nine tens and ten ones. Ones: ten take eight leaves two. Tens: nine take four leaves five. Hundreds: two take one leaves one.

152

Try it: Subtract $700 - 235$.

check: **465**

Skill 3 — Estimating a difference first

Rule: Round *both* numbers to the nearest hundred, then subtract the rounded ones.

Look at the tens digit: below five rounds down, five or more rounds up.

The estimate is a safety net. A regrouping that goes missing usually throws the answer out by a whole hundred, which the estimate catches at a glance.

Example: Estimate $604 - 287$ to the nearest hundred, then subtract exactly.

Step 1. Borrowing across a zero is where answers go wrong by a hundred, so I want a rough answer in hand before I start, to compare against afterwards.

Step 2. Rounding gives six hundred and three hundred, and their difference is the estimate. Exactly: open a hundred, leaving five hundreds and ten tens, then nine tens and fourteen ones. Ones: fourteen take seven leaves seven. Tens: nine take eight leaves one. Hundreds: five take two leaves three.

estimate **300** exact **317**

Try it: Estimate $902 - 458$ to the nearest hundred, then subtract exactly.

check: estimate 400, exact 444

Skill 4 — Finding a lost regrouping

Rule: The two classic slips are taking the smaller digit from the larger one in a column, and opening a hundred but forgetting to lower the hundreds digit.

To find either, redo the subtraction column by column and stop at the first digit that disagrees.

Example: Someone worked out $600 - 173$ and wrote 573. What went wrong?

Step 1. An answer this much too big is the signature of subtraction done backwards inside a column, so I redo the columns and look for the first one where the bigger digit was put on top.

Step 2. The correct regrouping is five hundreds, nine tens and ten ones. Ones: ten take three leaves seven, not the three that was written — the digits were flipped. Tens: nine take seven leaves two. Hundreds: five take one leaves four.

correct difference **427**

Try it: Someone worked out $500 - 264$ and wrote 364. Find the correct difference.

check: 937